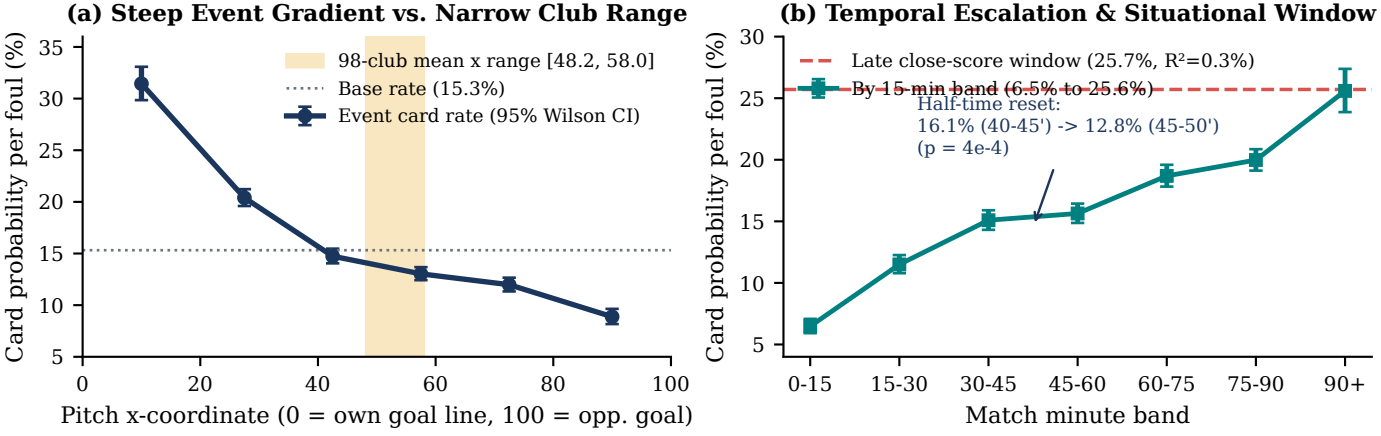
(a) Steep Event Gradient vs. Narrow Club Range
98-club mean x range [48.2, 58.0]
Base rate (15.3%)
Event card rate (95% Wilson CI)
Card probability per foul (%)
Pitch x-coordinate (0 = own goal line, 100 = opp. goal)
(b) Temporal Escalation & Situational Window
Late close-score window (25.7%, R²=0.3%)
By-15-min band (6.5% to 25.6%)
Half-time reset:
16.1% (40-45') -> 12.8% (45-50')
(p = 4e-4)
Card probability per foul (%)
0-15
15-30
30-45
45-60
60-75
75-90
90+
Match minute band